



COUNTING FROM THE FIRST TICK

# Life, the Universe, and God

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*A Software Engineer's Instrument for the Immeasurable*

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2026 · built against the ucal source tree

# Contents

|                                              |    |
|----------------------------------------------|----|
| The rigor is the medium .....                | 1  |
| What this is not .....                       | 2  |
| Two paths through the book .....             | 2  |
| A word about being wrong .....               | 3  |
| Part I — Foundations .....                   | 5  |
| 1 — Intent .....                             | 6  |
| The irritation, stated precisely .....       | 7  |
| What would have to be true instead .....     | 7  |
| What was built .....                         | 8  |
| Where the trouble started .....              | 8  |
| What this book will and will not claim ..... | 9  |
| 2 — The tick .....                           | 11 |
| Why this unit .....                          | 12 |
| What the tick is not .....                   | 12 |
| The concession: the tick's length .....      | 13 |
| Counting, not measuring .....                | 14 |
| 3 — The zero of time .....                   | 16 |
| The domain is unsigned .....                 | 17 |
| Why it is stipulated and not measured .....  | 18 |
| What is actually claimed .....               | 18 |
| The datum is in ordinary company .....       | 19 |
| 4 — The universe second .....                | 21 |
| Why base five .....                          | 22 |
| The ladder .....                             | 22 |
| What the ladder buys .....                   | 23 |
| What the ladder costs .....                  | 24 |
| The mark .....                               | 25 |
| Part II — What was built .....               | 27 |
| 5 — The domain .....                         | 28 |
| Unsigned, and why that is not a saving ..... | 29 |
| Closed: 512 bits, and what that buys .....   | 29 |
| Checked: overflow is a typed error .....     | 30 |
| Two backends, one domain .....               | 31 |
| No floating point. Anywhere. ....            | 31 |

|                                                        |    |
|--------------------------------------------------------|----|
| The two widths .....                                   | 33 |
| 6 — Notation .....                                     | 34 |
| Two text forms, and why both .....                     | 35 |
| Where each form anchors .....                          | 36 |
| The canonical binary form .....                        | 36 |
| UCID: the transcribable form .....                     | 37 |
| Four forms, one integer .....                          | 39 |
| 7 — The bridge .....                                   | 40 |
| One constant .....                                     | 40 |
| Why the direction matters .....                        | 41 |
| Refusing rather than rounding .....                    | 42 |
| The invariants that fall out free .....                | 42 |
| One pivot, and where the leap seconds live .....       | 43 |
| What the bridge costs .....                            | 45 |
| 8 — Derived calendars .....                            | 47 |
| Four components .....                                  | 48 |
| Units, as rationals of ticks .....                     | 48 |
| Intercalation is derived, never declared .....         | 48 |
| Cycles, and the absence of them .....                  | 50 |
| The anchor: the one thing that cannot be derived ..... | 51 |
| Derived and legacy, held apart .....                   | 52 |
| Part III — What implementation refused .....           | 55 |
| 9 — Divergence .....                                   | 56 |
| 10 — The 97/400 correction .....                       | 58 |
| Part IV — The datum .....                              | 60 |
| 11 — Why it cannot be measured .....                   | 61 |
| 12 — What was done instead .....                       | 63 |
| 13 — Florensky's radius .....                          | 65 |
| Part V — Any celestial body .....                      | 67 |
| 14 — Universal baselines .....                         | 68 |
| 15 — Deriving a calendar .....                         | 70 |
| 16 — Where it breaks .....                             | 72 |
| 17 — The one qualification .....                       | 74 |
| Part VI — Nine readings .....                          | 76 |
| 18 — Greek .....                                       | 77 |
| 19 — Jewish .....                                      | 79 |
| 20 — Islamic .....                                     | 81 |

|                                                  |     |
|--------------------------------------------------|-----|
| 21 – Patristic and Latin .....                   | 83  |
| 22 – Orthodox .....                              | 85  |
| 23 – Latter-day Saint .....                      | 87  |
| 24 – Modern philosophy .....                     | 89  |
| 25 – Russian .....                               | 91  |
| 26 – Method .....                                | 93  |
| Part VII – The instrument as research tool ..... | 95  |
| 27 – Six samples .....                           | 96  |
| 28 – Null results .....                          | 98  |
| Part VIII – The claim .....                      | 100 |
| 29 – Why a program .....                         | 101 |
| 30 – Uselessness restated .....                  | 103 |
| 31 – What is not claimed .....                   | 104 |
| 32 – Kant's moon .....                           | 105 |
| About the Author .....                           | 107 |
| A work of love .....                             | 108 |
| A note on cooperation .....                      | 108 |
| Where to find more .....                         | 108 |

## Before we begin

Three things are true about this project, and the third only follows if you accept the first two. So they go here, on the first page, before anything has a chance to soften them.

**The artifact is real.** `ucal` is working Rust — six crates, a library and a command line, published and installable. Time is an unsigned integer count of Planck-time units since a stipulated datum. There is no floating-point value anywhere in the workspace, not in a signature, a field, an intermediate, or the rendering path. Every constant in it is reproducible by two independent derivations that agree bit for bit. At the commit this book is pinned to, 381 tests pass on both integer backends. You can check every sentence in that paragraph; the last page of this book tells you how.

**Its practical utility is questionable.** No task you have today needs a Planck-tick count since a stipulated datum. It does not compete with `chrono`, `time`, or `hifitime`, and it does not try to. Nothing on its ladder of units is near a second or an hour — a second is 21.4 beats, an hour is 24.6 arcs — so it is not merely unnecessary for ordinary work but actively awkward for it. There is exactly one qualification to this, it concerns timekeeping across more than one planet, it takes up a single page in Part V, and it is not a rescue.

**Therefore it is research of another kind.** Art, philosophy, theology — conducted in the medium of a working program. That is not a consolation prize awarded after usefulness failed. It is what the thing is.

## The rigor is the medium

Here is the part that matters, and it is easy to misread as modesty.

That the code compiles, that the constants reproduce, that the tests pass — these are not preliminaries to the argument. They **are** the argument, in the same way that the metre of a poem is not preliminary to the poem.

An essay can assert that a distinction ought to be respected. It can argue at length, and persuade you, and then it is over, and the discipline it recommended

survives exactly as long as the next author's attention does. A type system can do something an essay cannot: it can make violating the distinction **fail to build**. Not discouraged. Not deprecated with a warning. Refused, by a machine, to a person who disagrees.

That difference is the thesis of this book, and the reason it is a book about a program rather than a program with a book attached.

## What this is not

Some of these will be obvious. They are listed because a book that surveys nine philosophical and religious traditions alongside a Rust crate will be misread in predictable directions, and it is cheaper to close those doors now than to keep apologising later.

**Not an apologetic** No tradition is argued true here. The code is evidence for none of them. Where a tradition and the instrument agree, that is reported as a convergence and never as a vindication of either.

**Not a proof of usefulness** The second fact above stands unretracted.

**Not a claim of discovery** When a tenth-century argument turns out to match a twenty-first-century type signature, the older text is not thereby shown to have anticipated anything. Convergence is not anticipation.

**Not numerology** No constant, base, or magnitude in this system acquires meaning by resembling a number in a tradition. Base 5 was chosen because  $5^5 = 3125$  gives a tier of exactly five digits. That is the whole reason. The book says so plainly and then reports, as a curiosity and not as evidence, that the first place the research went looking for a five, it found one waiting.

**Not a tutorial** If you want to use the crate, its documentation is better at that than this book will be.

**Not a defence of the datum** Tick zero is stipulated. That it is stipulated is the point, and Part IV is four chapters about why.

## Two paths through the book

The book serves two readers who are rarely the same person, and it does not ask either of them to pretend to be the other.

The **engineer's path** is Parts I, II, III, and V. It is a complete technical article about an unusual piece of software: what it computes, what its type system refuses, where the design lost arguments with the compiler, and how far the approach generalises beyond Earth. You can stop at the end of Part V and have received the whole of that.

The **reader's path** is Parts I, IV, VI, VII, and VIII. It is a book about what it means to build a measuring instrument that points at something it declares itself unable to measure, and about nine traditions that have thought carefully about origins, number, and time — read here as readers of the artifact rather than as authorities over it.

Part I belongs to both. Nothing later in the book may redefine what Part I establishes.

#### A NOTE ON MARKING

This book marks its own claims. Where a passage is **interpretation** — the author reading a meaning into the artifact — it sits in a ruled block that says so. Where a coincidence is genuinely striking but proves nothing, it is labelled **resonance**, and it may never appear as a premise for anything.

That apparatus is not decoration. It is the book submitting to the discipline it documents: the software marks where it stops computing, and the book marks where it stops asserting fact. If you delete every marked block, every technical claim in Parts I through III and V must still stand. Making sure that is true is a scheduled step in producing this book, not a hope about it.

## A word about being wrong

The most valuable document in this project's history is a correction.

Two revisions of the specification asserted that the continued-fraction machinery reproduces the Julian and Gregorian calendar rules as convergents. The Julian rule does appear — it is the first convergent. The Gregorian rule does not appear at any depth, and a rule twelve times simpler is more accurate than it. The machinery contradicted its author, in writing, on a claim he had published twice.

That correction is in this book, dated, including the revisions that were wrong. It is in here because a system built to make its author's conclusions provable would

never have produced it, and because the charge that such systems are **always** built that way is nine hundred years old and deserves an answer made of evidence rather than assurance.

If you find yourself reading a passage that seems to be presenting that correction as foresight rather than as an error published — a demonstration of rigour rather than an argument the author lost — then the book has failed at the thing it cares most about, and you should say so.

PART I

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# Foundations

*Definitional. Nothing later in the book may redefine these,  
and a reader who stops here must not leave with a false  
impression.*

CHAPTER 1

# 1

## Intent

---

This began as an irritation, and it is worth being honest that it began as an engineer's irritation rather than a philosopher's question.

Read almost any account of the early universe and you will find a sentence like this one: **recombination occurred about 380,000 years after the Big Bang**. It is a good sentence. It is doing real work. And if you stop and ask what its units are, something uncomfortable happens.

A year is the time Earth takes to go around the Sun. Not approximately — definitionally. The Julian year used in astronomy is exactly 365.25 days of exactly 86,400 seconds, and those numbers are what they are because of how fast one particular

rock spins and how long it takes to circle one particular star. So **380,000 years after the Big Bang** means  $380,000 \times 365.25 \times 86,400$  seconds, which means it is a quantity expressed in units defined by the rotation and orbit of a planet that would not exist for another nine billion years.

The number is correct. It is not wrong in any way that matters to a cosmologist. But it carries a passenger, and the passenger is Earth.

## The irritation, stated precisely

The problem is not that the units are arbitrary. All units are arbitrary. The metre is a bar in a vault, then a wavelength, then a fraction of a light-second; none of that makes it a bad metre.

The problem is **provenance leaking into arithmetic**. When you compute with Earth years, Earth's orbital period is inside every intermediate value. Usually that costs nothing. Occasionally it costs you a rounding you did not intend, because 365.25 days is not a whole number of anything and the conversions do not close. And structurally — which is the part that would not let go — it means the description of an event 13.8 billion years old is phrased in terms of a body that had no bearing on it.

### WHAT THIS IS NOT AN ARGUMENT FOR

None of this is a criticism of how cosmology is done. Astronomers use Julian years because they are stable, agreed, and unambiguous, which is exactly what a unit should be. The complaint here is narrow and it is aesthetic before it is technical: a quantity ought to be expressible in units that do not smuggle in a planet.

## What would have to be true instead

Suppose you wanted a time system with no Earth content in its arithmetic. What would it need?

It would need a **unit** that is not defined by any body's motion — something built out of constants of nature rather than out of a rotation.

It would need an **origin** that is not a calendar event, since every calendar epoch is someone's civil history.

It would need a **notation** in which writing fewer digits means saying less precisely, rather than meaning zero — because at these magnitudes you will constantly be quoting numbers you do not know to full precision, and a system that silently pads them with zeros is lying on your behalf.

And it would need to be honest about the one thing it cannot do: it cannot tell you what time it is on Earth without, at some point, being told about Earth. That contact has to happen somewhere. The question is whether it happens **in the arithmetic** or at a declared boundary you can point to.

That last question turned out to be the whole design.

## What was built

**ucal** is the answer to those four requirements, and Part II is the technical account of it. In one paragraph: absolute time is an unsigned integer count of Planck-time units since a stipulated datum; those ticks are grouped in a ladder of powers of five, so a timestamp is the tick count written in base 5 and truncating it **is** rounding it; Earth enters through exactly one declared constant and leaves at the formatting boundary; and no floating-point value appears anywhere in the workspace.

It works. It is also, as the preface said and will keep saying, of questionable use.

## Where the trouble started

The requirement that has no clean answer is the second one: the origin.

If time is a count since some zero, the zero has to be somewhere. And the obvious place to put it — the beginning — is not available, for reasons that took four chapters of this book to state properly and that Part IV is entirely about.

The short version, so you can carry it through Parts II and III: **the age of the universe is a measurement, and measurements have error bars, and an error bar cannot be the origin of an exact integer count.** The published figure is 13.787 billion years give or take 0.020 billion. That uncertainty, converted to ticks, is a number with fifty-eight digits. If the datum inherited it, every timestamp in the

system would inherit it too, and the exactness the whole design exists to protect would be theatre.

So the datum is **stipulated**. Declared, not discovered. Tick zero is a reference point that the specification says, in as many words, is not a measurement and not an observed event.

And then the interesting problem: how do you say that in a way that a future maintainer — or the author, three years later, in a hurry — cannot quietly forget?

#### INTERPRETATION

The answer this project arrived at is the reason it turned into a book. The physical claim about the origin is declared, cited, and given an exact magnitude. And then it is made **impossible to compute with**: a type with no arithmetic operations at all, guarded by a test that fails to compile if the type ever reaches an operand position.

You cannot forget a discipline that the compiler enforces. That is a small technical fact with a large philosophical shadow, and the shadow is what Parts IV, VII, and VIII are about.

## What this book will and will not claim

Before Part II starts and the prose becomes technical for a hundred pages, here is the calibration.

It **will** claim that a distinction enforced in a type system is a philosophical contribution and not merely an implementation detail. It will claim that the approach generalises to any celestial body, and it will spend a chapter on exactly where it fails. It will claim that the instrument does philosophical work no prose argument can do, and it will demonstrate that rather than assert it.

It will **not** claim that time began, that any tradition is correct, that the base is meaningful, or that anyone should adopt this calendar.

#### WHAT THIS CHAPTER ESTABLISHED

- Cosmic durations are conventionally expressed in units defined by one planet's motion — correct, useful, and carrying a passenger.
- A time system with no Earth content in its arithmetic needs a physical unit, a non-calendrical origin, a notation where truncation means imprecision, and one declared point of contact with Earth.
- The origin cannot be measured, because a measurement has error bars and an exact integer count cannot inherit one.
- So it is stipulated — and the project's contribution is making that stipulation mechanically impossible to forget.

CHAPTER 2

# 2

## The tick

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Everything this system computes is an unsigned integer count of ticks. Nothing else is primitive — not the second, not the beat, not the day. If you understand why the tick was chosen and, more importantly, what choosing it does **not** claim, you have the foundation of the whole instrument.

**TERM Tick**

The Planck time,  $t_P = \sqrt{\hbar G/c^5} \approx 5.391 \times 10^{-44}$  seconds. In this system it is the atomic unit of duration: the smallest quantity representable, and the unit in which every other quantity is counted.

**Why this unit**

The Planck time is composed from three constants: the gravitational constant  $G$ , the reduced Planck constant  $\hbar$ , and the speed of light  $c$ . Those three bound gravitation, quantum action, and the propagation of causality respectively — and there is exactly one combination of them with the dimension of time.

That is the appeal, and it is worth stating carefully because it is easy to overstate. The tick is not defined by any body's motion. It does not know about Earth, or about the Solar System, or about matter being organised into planets at all. It is the interval you get when you ask the three limiting constants of physics what a duration would be, and they answer.

For a system whose entire purpose is to have no Earth content in its arithmetic, that is the right floor.

**What the tick is not**

Here is where a great deal of writing about the Planck time goes wrong, and where this project declines to follow.

**The tick is not a quantum of time.** It is the resolution floor of an instrument, not a structure discovered in the world. Nothing in this project asserts that time comes in discrete lumps, that there is a smallest possible interval, or that asking about a shorter duration is physically meaningless. The system cannot represent a shorter duration; that is a fact about the system.

The distinction matters more than it looks, because asserting otherwise would silently take a side in an argument that has been running for twenty-four centuries.

Aristotle holds in **Physics** VI that time is a continuum: divisible without limit, and the instant is a **limit** of an interval rather than a part of it, in the way that a

point is a limit of a line and not a very short line. On this view a smallest duration is not merely unobserved but incoherent.

Against that, temporal atomism has a long and respectable history. Epicurus held that there are **minima** of time as there are minima of magnitude. The Kalām tradition developed a thoroughgoing atomism in which time is composed of indivisible moments and accidents do not endure from one to the next.

The instrument is discrete. If discreteness were asserted as a claim about the world, this project would have joined Epicurus and the Kalām against Aristotle — and it would have done so not on the strength of an argument but as a side effect of choosing an integer type.

That is not a respectable way to take a metaphysical position. So the position is declined, explicitly, and the declining is written down: the tick is where the instrument's resolution stops, and the instrument makes no claim about where the world's does.

#### A RULE YOU WILL SEE AGAIN

This pattern — **the implementation forces a choice; name the choice; refuse the claim that would come free with it** — recurs throughout the design. It is the same move that Part IV makes about the datum, at much greater length and with much higher stakes.

## The concession: the tick's length

There is one place where the tick is not free of Earth, and it should be admitted here rather than discovered later.

The Planck time's **numerical value** is known only as well as  $G$  is measured, and  $G$  is the worst-measured of the fundamental constants by a wide margin. Worse, expressing it at all requires a unit of time to express it **in** — and that unit is the SI second, which is defined by a caesium transition counted by instruments on Earth.

So the tick's length is fixed by convention against the second. The system declares a value, records where the value came from, and uses that declared value everywhere. It does not re-derive it, and it does not pretend the value is more certain than the measurement behind it.

## INTERPRETATION

What this concedes is **metrology** and nothing else. The arithmetic contains no Earth content: ticks are counted, not converted. What is Earth-flavoured is the statement of how long a tick is in seconds — which is a sentence about translating between two systems, not a fact used inside either.

Whether that is a satisfying answer is a fair question, and Part III returns to it. It is at least an honest one, and its honesty is structural: the conversion lives at a declared boundary you can point to rather than dissolved into the operations.

## Counting, not measuring

One more consequence, which will matter in Part IV.

Because time here is a **count**, the system's exactness is the exactness of integer arithmetic. Tick 8,070,204,002,895,596,515,944,343,085,635,637,180,530,466,139,316,558,837,890,625 is a specific tick, and adding one to it gives the next one, and that is all perfectly exact — in the same way that the number of steps you have walked is exact whether or not you know it.

What is **not** exact, and cannot be made exact by any amount of careful arithmetic, is the correspondence between tick zero and any physical event. That is a separate question, it has a separate answer, and keeping those two things apart is the discipline the rest of this book is about.

#### WHAT THIS CHAPTER ESTABLISHED

- The tick is the Planck time, built from  $G$ ,  $\hbar$ , and  $c$  — the one combination of the three limiting constants with the dimension of time.
- It is the resolution floor of an instrument, **not** a quantum of time. Asserting otherwise would take a side in a live argument as a side effect of choosing an integer type.
- Its length in seconds is fixed by convention against SI. This concedes metrology and nothing else, at a boundary you can point to.
- Time is counted, not measured. The count is exact; what tick zero corresponds to is a different question entirely.

CHAPTER 3

# 3

## The zero of time

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A count needs somewhere to start counting from. This chapter says where, and says the one thing about it you need in order to read Parts II and III without being misled. The full argument is Part IV, four chapters of it, and the deferral is deliberate — but a reader who stops at the end of Part I must not leave with a false impression, so the essential point is made here and made plainly.

**TERM Datum**

Tick zero. A **stipulated** reference point, conventionally identified with the Friedmann–Lemaître–Robertson–Walker  $t \rightarrow 0$  limit. It is not a measurement and not an observed event.

## The domain is unsigned

Absolute time in this system is an unsigned integer. There is no tick  $-1$ .

This is not a storage optimisation. It is a statement about what the system is willing to represent: nothing precedes the datum, because the datum is where the count begins, and a count does not go backwards past its own start.

Ask for an instant before tick zero and you do not get a negative number. You get an error — **UCAL-E0020** — and the operation fails.

**INTERPRETATION**

There is a difference between answering a question with a negative number and refusing the question, and the system takes the second option deliberately.

A negative tick count would be an **answer**: it would say that the time before the datum exists, is measurable, and happens to lie on the other side of a chosen origin — the way 1 BC lies on the other side of the Gregorian epoch. The refusal says something else: that the question, as posed to this instrument, is malformed.

Augustine gives the classic form of the move in **Confessions** XI, answering what God was doing before creating heaven and earth. His answer is not a description of that interval but a rejection of the question’s shape, because time is among the things created and there is no “before” of the sort the question wants. Whether or not you find the theology congenial, the logical structure is exactly **UCAL-E0020**: an error, not a negative number.

## Why it is stipulated and not measured

Here is the one-line version. Part IV gives it four chapters and three further reasons; this is the one you can carry.

## Exactness cannot come from measurement.

The age of the universe is a measured quantity. The current best figure is 13.787 billion years, plus or minus 0.020 billion. That is an excellent measurement. It is also, like every measurement, an interval rather than a point.

Convert that uncertainty into ticks and it becomes:

- ● ● *ucal datum — the claim's magnitude*

[illegible]

Fifty-eight digits. About 0.145% of the whole span from the datum to now.

Now suppose the datum were defined **as** the measured age. Every timestamp in the system would inherit that uncertainty, because every timestamp is an offset from the datum. The exact integer arithmetic — the thing this entire design exists to protect — would be computing precise-looking answers on top of a foundation that wobbles by a number with fifty-eight digits in it.

You cannot get an exact origin from an inexact measurement. So the origin is not taken from the measurement. It is **declared**, and the measurement is recorded separately.

## What is actually claimed

Precision about this matters, so here is the claim in full, with nothing left implicit.

The system asserts that tick zero is tick zero. That is a definition, and definitions are not the sort of thing that can be wrong.

It asserts, **separately and as metadata**, that tick zero is conventionally identified with the FLRW  $t \rightarrow 0$  limit, and that the published uncertainty in that identification is  $\pm 0.020$  Gyr, citing Planck 2018. This is a claim about the world and it may be revised.

And it asserts nothing whatever about whether time began, whether there was a first moment, or whether the FLRW limit corresponds to a physical event at all.

#### DEFERRAL IS NOT EVASION

You have now been told that the datum is a decision rather than a discovery, and why the short reason forces it. What Part IV adds is three further reasons the specification gives, one more that it does not, and — the part that turned this project into a book — what was **done** about it: how a claim can be declared, cited, given an exact magnitude, and simultaneously made impossible to compute with.

If you are on the engineer’s path and skipping Part IV, the sentence to carry forward is this one: the uncertainty in the age of the universe does not make the timestamps uncertain, because it is not an operand.

## The datum is in ordinary company

One last thing, because “stipulated origin” can sound exotic and is not.

TAI’s zero is 1 January 1958, chosen by agreement. The Julian Day epoch is 1 January 4713 BC, chosen because it made a convenient common multiple of three cycles. The Unix epoch is 1 January 1970, chosen because it was recent and round. None of these is a discovery about the universe; all of them are decisions that turned out to be useful.

#### INTERPRETATION

What is unusual here is not that the origin is stipulated. It is that the stipulation is **declared as such in the artifact**, with the physical claim held separately and marked non-computable — rather than being a fact about the system’s history that you would have to go and read about.

Most epochs are stipulated and silent about it. This one is stipulated and says so, in a place the compiler can reach.

#### WHAT THIS CHAPTER ESTABLISHED

- Tick zero is the datum: stipulated, conventionally identified with the FLRW  $t \rightarrow 0$  limit, and explicitly not a measurement.
- The domain is unsigned. A request for a time before the datum is an error, not a negative number — a refusal of the question rather than an answer to it.
- Exactness cannot come from measurement: the  $\pm 0.020$  Gyr uncertainty is a 58-digit number of ticks, and an exact count cannot inherit it.
- The physical claim is recorded separately as metadata, cited, and — as Part IV shows — made impossible to use in arithmetic.

CHAPTER 4

# 4

## The universe second

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A tick is far too small to think in. It is  $5.4 \times 10^{-44}$  seconds; the age of the universe is about  $8 \times 10^{60}$  of them. Nobody reads a 61-digit number.

So there is a ladder of larger units above the tick, and it has an unusual property: every rung is a whole power of five of the one below, and the whole ladder is a whole power of five of the tick. That property is what makes the notation work, and this chapter is about what it buys and what it costs.

**TERM Beat**

$5^{60}$  ticks, about 46.762 milliseconds. The reference rung of the ladder, and what the specification calls the **universe second** — a unit of human-noticeable size with no Earth content whatever.

**Why base five**

Because  $5^5 = 3125$ , and 3125 is a number of exactly five base-5 digits.

That is the entire reason. It is worth being blunt about it, because a book that spends its second half among Pythagoreans and Neoplatonists has an obligation to say clearly where its numbers came from, and this one came from a digit-packing convenience.

**RULE N**

No constant, base, or magnitude in this system acquires significance by resembling a number in a tradition. Not the five, not the sixty in  $5^{60}$ , not the 3125.

The research did, as it happens, go looking. The first place it looked — Losev on the Neoplatonic tetrad — had a five waiting, and the Pythagorean tetraktys sums to ten and builds from four. Those are interesting facts about those texts. They are not evidence about this software, they were not inputs to it, and the book reports them once, in Part VI, labelled as what they are.

**The ladder**

Each rung is  $5^5 = 3125$  of the rung below, which is to say each rung is exactly five base-5 digits wide. Rung  $k$  is  $5^{60+5k}$  ticks, indexed from the beat at  $k = 0$ .

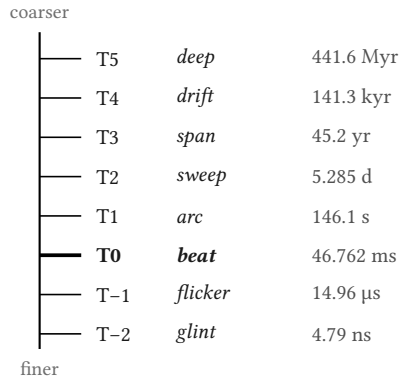


Figure 2 — The named rungs of the tier ladder. Each is 3125 of the one below. The ladder continues in both directions; only the named tiers are shown.

## What the ladder buys

Three things, and they are all consequences of the same fact — that a timestamp is the tick count written in base 5 and grouped in fives.

**Truncation is rounding.** Write fewer groups and you have said the same thing less precisely. There is no separate rounding step, no scaling, no loss of a different kind: the digits you dropped are exactly the precision you gave up. This is the property that makes the uncertainty discipline of Part II possible at all.

**Prefix comparison is chronological comparison.** Two timestamps compare in the order their digits compare, because base-5 positional notation is monotone. Sorting the text sorts the times.

**Writing all the digits pinpoints one tick.** There is no accumulated conversion error between the coarse view and the fine one, because they are the same integer read at different widths.

Here is one real instant, at full precision and then at three coarser tiers:

• • • *ucal explain — one instant, four precisions*

```

ticks
8070205189128471254993117657693008777530466139316558837890625

human      UC1 0031·0687·2481·3000·1638·3018:0779·2671·2006·1837·...
tiers:
  T5 deep   31
  T4 drift  687
  T3 span   2481
  T2 sweep  3000
  T1 arc    1638
  T0 beat   3018

```

The **0031** at the front is deeps since the datum. The **3018** before the colon is beats within the current arc. Every group is a base-5 number written in decimal, which is why none of them exceeds 3124.

#### INTERPRETATION

There is something worth noticing about that colon. It sits at the beat — at  $5^{60}$  — and it is the only place in the notation where a human convenience has been allowed in. It is there because the beat is roughly the scale at which a person can notice a duration, so it is the natural place to put the decimal point of a time system built for beings who perceive at that scale.

That is a concession to the reader, not to Earth. It changes no arithmetic.

## What the ladder costs

Nothing on this ladder is near anything you know.

A second is 21.385 beats — not a whole number, and not close to one. An hour is 24.6 arcs. A day is 0.189 sweeps. A year is 0.699 spans.

### THE TWO SECONDS DO NOT DIVIDE

This is the sharpest form of the cost, and the system says so on its own ladder output:

The two seconds are incommensurable above T-6. One bridge second is 21.385061835 beats, not a whole number, because **BEAT** carries  $5^{60}$  while **SECOND** carries only  $5^{30}$ . They share a common measure only at the tick.

So the beat is not a second in disguise, and no amount of rescaling would make it one. They agree at the tick and nowhere above it — which is exactly why the tick is primitive rather than either of them.

That is the honest consequence of leaving the Earth paradigm. If you want units with no planetary content, you do not get to keep the hour. The hour **is** planetary content.

## The mark

The project's own mark is a diagram of this chapter, and it is worth reading as one.

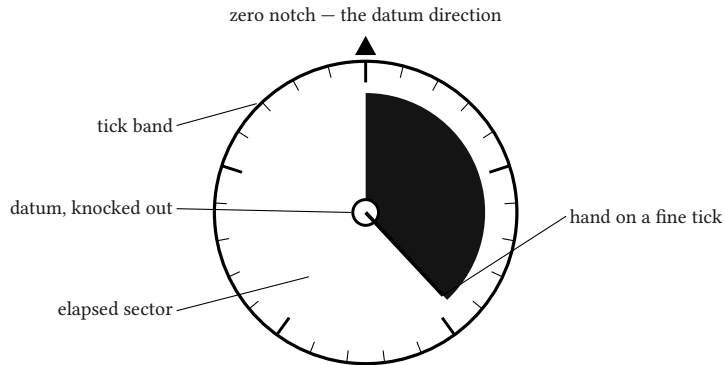


Figure 4 — The mark, read as a diagram. Five heavy divisions on the tick band, not twelve — the base is visible at a glance. The centre is negative space.

The centre dot has its core knocked out. That is deliberate: the datum is stipulated rather than observed, so the mark declines to draw anything there. The hand lands on a fine tick rather than a coarse division, because the claim the instrument makes is that it can pinpoint an arbitrary exact instant — not that it can round to a tidy one.

## INTERPRETATION

A mark that showed a filled centre would be making a claim the specification refuses. The identity was drawn after the rule existed, and it obeys it. That is a small thing, and it is the sort of small thing this project is made of.

## WHAT THIS CHAPTER ESTABLISHED

- The beat is  $5^{60}$  ticks  $\approx 46.762$  ms — the **universe second**, a human-scale unit with no Earth content.
- Base five because  $5^5 = 3125$  is five base-5 digits wide. Rule N: no further meaning, and the book says so before Part VI can be misread.
- The uniform ladder makes truncation **be** rounding, prefix comparison **be** chronological comparison, and full precision pinpoint one tick.
- The cost is that nothing on the ladder is near a second or an hour — one second is 21.385 beats, and the two share a measure only at the tick.

PART II

---

# What was built

*An engineer may stop at the end of this part and have  
received a complete technical article.*

CHAPTER 5

# 5

## The domain

---

Part I established what is counted. This part is about how the counting is implemented, and it starts with the number that holds the count.

Three properties, and the third is the one that turns out to have consequences all the way into the cosmology: the domain is **unsigned**, it is **closed**, and every operation on it is **checked**.

## Unsigned, and why that is not a saving

There is no tick  $-1$ . Chapter 3 said why in one line — nothing precedes the datum, because the datum is where the count begins — and the type system is where that line is enforced.

`Instant` has no `Sub` implementation returning a signed value. Subtracting a later instant from an earlier one does not produce a negative `Delta`; it produces `UCAL-E0020` and the operation fails. To get the distance between two instants regardless of order you must ask for it by name, and the name says what you are doing.

### THE SAME DISCIPLINE, APPLIED TO RATIONALS

`Ratio` — the exact rational type used throughout the derivations — carries the identical rule. `sub` refuses a negative result and directs the caller to `abs_diff`. That is Rule Z applied one level down, and it means the same class of mistake is impossible in the calendar derivations as in the timestamps.

## Closed: 512 bits, and what that buys

The tick count is a 512-bit unsigned integer. The ceiling is  $2^{512} - 1$ , which the system will print for you:

● ● ● *ucal doctor — the domain*

```
domain_max_ticks
13407807929942597099574024998205846127479365820592393377
72356144372176403007354697680187429816690342769003185818
6486050853753882811946569946433649006084095
domain_bits      512
```

In years, that ceiling is about  $2.29 \times 10^{103}$ .

To give that a scale: proton decay, if it happens at all, is expected somewhere around  $10^{34}$  years. The last black holes evaporate around  $10^{100}$ . The domain outlasts both, with room left.

The present epoch sits at about  $8.07 \times 10^{60}$  ticks — which is  $6.0 \times 10^{-94}$  of the range. The counter is, for all practical purposes, still at zero.

#### A NUMBER THIS BOOK HAD TO CORRECT

RFC UCAL-A1 attaches the figure  $7 \times 10^{-17}$  to this paragraph. That number is real but belongs to a different range: it is the present epoch as a fraction of the **UCID's** 256-bit space, not of the 512-bit domain. The domain fraction is  $6.0 \times 10^{-94}$ .

Rule S is why this correction is here rather than the original figure. The article RFC describes what the book intended to say; the source tree is what is true. This is the first place in the book where those two disagreed, and it will not be the last.

#### INTERPRETATION

The width was not chosen to be impressive. It was chosen so that **the width never has to change**, because under Rule B the value width is a wire-format commitment: the canonical binary encoding is 64 bytes because the domain is 512 bits, and widening the domain later would invalidate every stored timestamp in existence.

A system that might need to grow its integer has a migration in its future. This one bought its way out of that with sixty-four bytes, once.

## Checked: overflow is a typed error

No operation on a time type wraps, and none saturates.

Every arithmetic method that could exceed the domain returns a **Result**. Overflow is **UCAL-E0060**, an ordinary error you handle, not a silently wrong answer you ship. The release profile sets **overflow-checks = true**, so the guarantee

holds in optimised builds and not only in debug ones — which is where this class of bug normally hides.

There is also a workspace lint that fails the build on `wrapping_*` or `saturating_*` appearing on a time type. That is belt and braces, and it is there because the failure mode it guards is the quietest one available: a wrapped tick count is not an error message, it is a timestamp from the wrong end of eternity that looks entirely plausible.

## Two backends, one domain

The tick integer has two implementations. The default is a fixed 512-bit stack integer; the alternative is an arbitrary-precision heap integer. They are selected by feature flag and are mutually exclusive at compile time.

The interesting requirement is that they must accept and reject **exactly the same values**. The arbitrary-precision backend is unbounded and would happily represent  $2^{513}$ ; it is made to enforce the same ceiling anyway, and the whole test suite runs against both.

### INTERPRETATION

It would have been easier to let the unbounded backend be unbounded. The reason not to is that the two would then be different systems wearing the same name — a timestamp accepted by one and rejected by the other, with no way to tell from the type which you were holding.

The rule that forbids this is one line long, and enforcing it cost a full second copy of the test suite in CI. That ratio — a cheap sentence, an expensive guarantee — is characteristic of the whole design.

## No floating point. Anywhere.

This is the constraint with the largest blast radius, so it gets stated in its strongest form: there is no floating-point value in any shipped crate of this workspace. Not in a signature, not in a field, not in a constant, not in an intermediate, and not in the rendering path that produces the human-readable output.

A lint scans the source and fails the build on any float token. It permits exactly one exception — a float reference implementation used as a **test oracle** — and it requires the exception to be marked, and it prints every exemption it honoured on every run, so an escape hatch cannot quietly become a retreat.

### What it costs

The cost is real and it is concentrated in one place: cosmology.

Computing the age of the universe at a given redshift means evaluating an integral. In any normal library that is a call to a quadrature routine over `f64`, and it returns a number with an error estimate. Here there is no `f64`, so it is certified interval quadrature over exact rationals — thousands of panels, each bounded above and below, with directed integer square roots underneath.

It is slower by orders of magnitude. The default subdivision depth takes about half a second where a float routine would take microseconds.

### What it buys

An **enclosure** rather than an estimate.

The float routine returns a number and a tolerance, and the tolerance is a well-informed guess about accumulated error. The interval routine returns two numbers and a proof that the true value lies between them. Not probably. Provably, given the model.

## INTERPRETATION

For most software the float answer is the better engineering, and it is not close. A guess accurate to twelve digits, computed instantly, beats a proof accurate to four, computed slowly.

This project takes the other option because its subject is a quantity whose uncertainty is the interesting part. Part IV is four chapters about an origin that cannot be measured; a cosmology module that reported a confident point estimate would be undermining the book's central claim in the one place where the claim is most testable.

## The two widths

One consequence deserves its own note, because it is where the arithmetic discipline becomes an argument.

Every cosmological result carries **two** uncertainty widths, and they are never merged: the width contributed by the quadrature, and the width contributed by the model’s own measured parameters. Asking for a single combined tolerance requires calling a method with a name that says you are combining them.

At recombination the quadrature width is about 251 years and the parameter width is about 10,900 years. Merged into one number, the answer would be “about eleven thousand years of uncertainty” — and the fact that forty-three forty-fourths of it comes from the **measurement** rather than from the computation would be invisible.

That distinction is what tells you more computing power would buy nothing here. A merged tolerance hides it. Keeping them apart is a line in the specification, and it is the reason the specification has that line.

### WHAT THIS CHAPTER ESTABLISHED

- The domain is unsigned: a result before the datum is `UCAL-E0020`, not a negative number. The same rule applies one level down, to `Ratio`.
- 512 bits reaches  $2.29 \times 10^{103}$  years, past proton decay and black-hole evaporation. The present epoch is  $6.0 \times 10^{-94}$  of it — and the width never has to change, because it is a wire-format commitment.
- Overflow is a typed error, enforced in release builds and by a lint, because a wrapped tick count looks entirely plausible.
- Two backends, one domain: the unbounded one enforces the same ceiling, verified by running the whole suite twice.
- No floating point anywhere. It costs orders of magnitude in the cosmology and buys a certified enclosure instead of an estimated tolerance.

CHAPTER 6

# 6

## Notation

---

One value, four ways of writing it down. This chapter is about why that is not redundancy, and about what each form is for.

• • • *ucal explain — one instant, four notations*

```

ticks
8070205189128471254993117657693008777530466139316558837890625

human      UC1 0031·0687·2481·3000·1638·3018:0779·2671·2006·1837·...

digit5     UC1/5 00000.00000. ... .00111.10222.34411.44000.23023. ...

ucid       0000000000050PM6K45MKCAVY5MPYAMHCJQ142JHE26A2ZAJ9FJ1

```

## Two text forms, and why both

The **human form** writes each tier group as a decimal number. The **digit form** writes the underlying base-5 digits directly, five to a group.

They encode the same integer. The difference is who is reading.

A person reading `0031·0687·2481` can see at a glance that this is 31 deeps, 687 drifts, 2481 spans — three quantities in a unit system, each written the way people write numbers. The same prefix in the digit form is `00111.10222.34411`, which is correct, canonical, and unreadable.

### INTERPRETATION

The temptation is to call the human form a convenience wrapper and the digit form the real one. That is not quite right, and the distinction matters for Part VIII.

The digit form is canonical **for machines** — parsing, sorting, round-tripping. The human form is canonical **for statements about time**, because tier groups are the units the system actually reasons in. Neither is a rendering of the other; both are renderings of the integer.

Each form carries a tag — `UC1` or `UC1/5` — so a value never travels without declaring which profile and which notation produced it. That is Rule P doing its work at the notation layer: two timestamps from different profiles cannot be compared by accident, because the text itself says they are different things.

## Where each form anchors

This is the detail that most repays care, and it is the subject of one of the corrections this project made against its own specification.

A form printed to fewer groups is **less precise**, not **padded with zeros**. Chapter 4 called this “truncation is rounding”. But saying that precisely requires knowing which tier each form counts from.

The human form anchors at **T0**, the beat: the group immediately before the colon. The digit form anchors at **T32**, the top of the ladder. So the same act of dropping trailing groups means the same thing in both — you have said less — but the tier a given group sits at is read from a different reference.

### D-A8 — AN AMENDMENT THE IMPLEMENTATION FORCED

The specification originally described the anchoring loosely enough that two readers could disagree about what a truncated form denoted. Implementation made the ambiguity concrete: the human form **cannot express a precision coarser than T0**, because it has no groups above the colon to drop.

That is not a defect. It is a consequence of anchoring at the beat, and it is why the command line falls back to a different form when asked to render a timeline at a tier above T0 — rather than printing a blank field, which is what the first implementation did and which looked exactly like a value.

## The canonical binary form

Sixty-four bytes, big-endian, fixed width. Never length-prefixed, never minimal, never a varint.

The width is not negotiable and the reason is worth stating: **byte order is numeric order**. A fixed-width big-endian encoding sorts lexicographically into chronological order, so the encoding is directly usable as a database key or a sort key with no comparator at all.

A minimal encoding — the obvious space optimisation — would break that, and it would break something else: the wire format would depend on the **magnitude**

of the value rather than on the profile. Two timestamps of different sizes would have different lengths, and a reader would have to know the value to know how to read it.

#### INTERPRETATION

Sixty-four bytes for a timestamp is extravagant by any ordinary standard. Unix time fits in eight.

What the extra fifty-six buy is that the format never has to change. The domain is 512 bits, the encoding is 512 bits, and there is no future in which a stored timestamp becomes unreadable because the range was widened. That is a real cost paid once against a class of migration that software normally pays repeatedly.

## UCID: the transcribable form

The fourth notation is a 52-character identifier in Crockford base-32.

• • • *the alphabet*

0123456789ABCDEFGHIJKLMNPQRSTUVWXYZ

**I**, **L** and **O** are absent because they are confusable with **1** and **0**. **U** is absent to avoid accidental obscenity. The alphabet is strictly ascending in ASCII — and there is a test that checks that rather than trusting it — which is what makes lexicographic order equal numeric order here too.

### Why 256 bits and not 512

Because the UCID is meant to be read, typed, and quoted by a person, and the domain is not.

At five bits per character, covering the full 512-bit domain would take **103 characters**. At 256 bits it takes 52, which is 260 bits of capacity — so the leading character encodes a single significant bit and is always **0** or **1**.

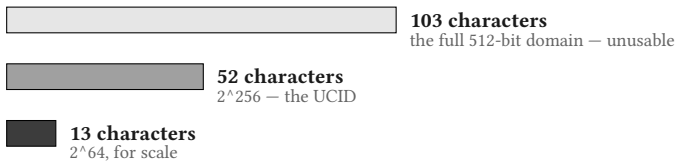


Figure 3 — UCID length against range. Five bits per character, so width is linear in bits — and the domain is twice the bits.

The ceiling that buys is  $2^{256}$  ticks, about  $1.98 \times 10^{26}$  years. That is past the end of the stelliferous era and thirty orders of magnitude beyond the present epoch — and far short of the domain’s  $2.29 \times 10^{103}$  years.

Above the ceiling you do not get a truncated identifier. You get `UCAL-E0031`, **instant outside UCID range**. The same refusal-rather-than-approximation that runs through the whole design.

### What UCID is not

The source documentation is emphatic here, and so is this book.

**UCID contains no randomness.** It is a pure function of the instant. Two events at the same tick receive the same identifier, which makes it useless as a unique key for concurrent events.

It is worse than merely non-random. Chapter 7 shows that an instant read from a nanosecond clock has at least twenty-one trailing base-5 zeros by construction, so the low-order characters are not just predictable but **structurally constrained** — consecutive nanoseconds share more than twenty leading characters.

There is a test called `ucid_has_no_entropy` whose entire job is to measure how badly the identifier would fail if someone used it as a UUID.

### INTERPRETATION

Writing a test that demonstrates your own type is unsuitable for an obvious misuse is an unusual thing to do. It costs nothing to omit; nobody would notice.

It is here because the failure would be silent. A UCID **looks** like a UUID — same length class, same character set, same shape in a log line — and the moment someone reaches for it as one, the collisions begin and never announce themselves. A test that fails loudly at the boundary is cheaper than an incident.

## Four forms, one integer

To close the chapter, the whole picture in one table.

| form                | for                                     | width    | sorts?         |
|---------------------|-----------------------------------------|----------|----------------|
| <code>ticks</code>  | the value itself, in decimal            | variable | no             |
| human               | statements about time, by people        | variable | only if padded |
| <code>digit5</code> | parsing, round-tripping, canonical text | variable | only if padded |
| binary              | storage, wire, database keys            | 64 bytes | yes            |
| UCID                | quoting, transcribing, URLs             | 52 chars | yes            |

The two that sort are the two that are fixed-width. That is not a coincidence, it is the whole mechanism, and it is why the text forms carry the caveat “only if padded” rather than being quietly assumed to sort.

### WHAT THIS CHAPTER ESTABLISHED

- Two text forms encode one integer: the human form in decimal tier groups, the digit form in raw base-5. Neither is a rendering of the other.
- Each form declares its anchor — human at T0, digit at T32 — which is what makes truncation mean **less precise** rather than **padded**.
- Canonical binary is 64 bytes, fixed-width, big-endian, so byte order is chronological order and the format never has to change.
- UCID is 52 characters over  $2^{256}$  because 512 bits would need 103. Beyond the ceiling is UCAL-E0031, not a truncation.
- UCID carries no entropy and is not a UUID. A test measures exactly how badly it would fail as one, because that failure would otherwise be silent.

CHAPTER 7

# 7

## The bridge

---

Chapter 1 said the design's whole question was whether Earth enters **the arithmetic** or at **a declared boundary you can point to**. This chapter is the boundary.

### **One constant**

Earth enters the system through exactly one number.

• • • *the bridge constant*

```
SECOND = 18 548 584 399 861 000 000 000 000 000 000 000 000 000
000 ticks
```

That is how many ticks are in one SI second. It is an exact integer — declared, not computed — and it is the only place in the workspace where a foreign unit is named at all. A lint fails the build if any identifier in the core crate mentions a foreign unit outside the bridge declaration.

#### INTERPRETATION

The number of things this constant is **not** doing is the point.

It is not a conversion factor applied throughout the code. It is not a scale on the tick. It does not appear in any arithmetic that produces a timestamp from another timestamp. Every operation the system performs on absolute time is integer addition, subtraction and comparison on tick counts, and none of them has ever heard of a second.

**SECOND** is consulted when a foreign value arrives and when one leaves. In between, it is inert.

## Why the direction matters

Converting **into** absolute time is multiplication by an integer. Multiplication of integers is exact. So a whole number of seconds becomes a whole number of ticks with no rounding whatsoever, ever, under any circumstances.

Converting **out** is division, which is where rounding lives — and so the rendering path is the only place a rounding mode is chosen, and it is always chosen explicitly by the caller.

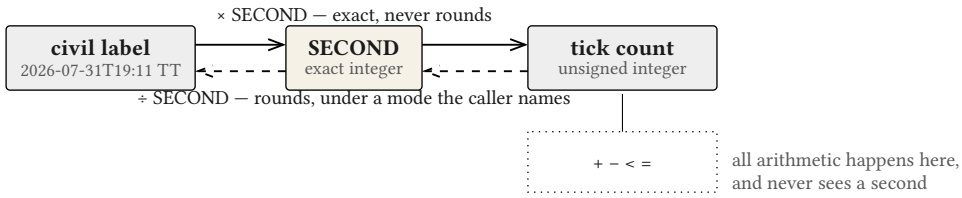


Figure 9 — The bridge. Earth enters and leaves at one declared constant; the arithmetic in between is integer operations on tick counts.

## Refusing rather than rounding

Ask the system to accept a time finer than the bridge constant can represent and it does not round. It returns **UCAL-E0043** — **foreign-unit input finer than the bridge constant permits**.

The threshold is  $10^{-30}$  seconds, and it is not arbitrary: it is where the declared constant's own decimal precision runs out. Below it, the system does not know the answer, and it says so instead of inventing one.

### THE PATTERN, AGAIN

This is the third time in three chapters. Before the datum: refuse the question (UCAL-E0020). Beyond the UCID range: refuse to truncate (UCAL-E0031). Finer than the bridge: refuse to round (UCAL-E0043).

A system that answered all three would be more convenient and would be lying in three different ways.

## The invariants that fall out free

Here is the part that makes the design feel less like a set of choices and more like a consequence.

**SECOND** was set to a multiple of  $10^{30}$ . Because  $10^{30} = 2^{30} \cdot 5^{30}$ , that means **SECOND** carries thirty factors of five. The nanosecond — a thousandth of a millionth of it — carries twenty-one.

So:

| quantity      | factors of 5 | consequence                                            |
|---------------|--------------|--------------------------------------------------------|
| SECOND        | 30           | a whole second lands with 30 trailing base-5 zeros     |
| NANOSECOND    | 21           | a whole nanosecond lands with 21 trailing base-5 zeros |
| ORIGIN_OFFSET | 61           | the bridge epoch is zero in every tier below T0        |

Read that in the notation of chapter 6 and it means something concrete: **any instant that came from a wall clock has a long tail of zeros in its digit form**. Six full tier groups of them, for a nanosecond clock.

#### INTERPRETATION

These are excellent invariant tests, and the project uses them as such — they are cheap to check and they fail loudly if the bridge is ever mis-declared.

They are also the reason chapter 6 could say the UCID is not merely non-random but **structurally constrained**. The zeros are not a coincidence about particular timestamps; they are a theorem about every timestamp that entered through the bridge. An identifier whose low-order digits are provably zero is not a candidate for a unique key, and it is better to know that from the invariant than to discover it from a collision.

## One pivot, and where the leap seconds live

Civil time is a mess, and the design's response is to confine the mess.

There is exactly one pivot scale: **Terrestrial Time**. TT is a uniform scale with no leap seconds, which is precisely why it was chosen — a timestamp converted through TT never has to know about the irregular ones.

Leap seconds exist, and the system knows all twenty-seven of them. They are applied at the parse and format boundary and nowhere else. No arithmetic on absolute time ever encounters one.

• • • *ucal doctor — the leap table*

```

leap_seconds:
  table_version      IERS Bulletin C 70 (no leap second to
2026-06-30)
  entries            27
  complete_through   2026-06-30
  pre_1972           the 1961-1972 rubber-second era is modelled
exactly;
                    UTC before 1961-01-01 is UCAL-E0041
  network            never; the table is bundled and offline

```

Two details in that output are worth pausing on.

**The table is bundled and offline.** No command in this system performs network access, ever. A time library that phones home is a time library whose answers depend on whether the network was up.

**The rubber-second era is modelled exactly.** Between 1961 and 1972, UTC did not tick at the same rate as TAI — it ran at declared fractional offsets that changed periodically, and the offsets were not whole seconds. Most software declines to handle this and starts its UTC support in 1972.

#### INTERPRETATION

It turns out to be exactly representable in this system, and the reason is a small arithmetic accident worth recording. The era's rate coefficients — 0.001296, 0.0011232, 0.002592 seconds per day — are all divisible by 27. A day is 86,400 seconds, which carries  $3^3 = 27$ . The 27s cancel, and what is left is exact in the rationals the system already uses.

Nobody designed that. It was discovered while implementing, and it meant an era that could have been declared out of scope became eleven years of exactly convertible history instead. Before 1961 there is nothing to convert exactly, and the system returns `UCAL-E0041` rather than guessing.

## What the bridge costs

Honesty, since Part III is about what implementation refused.

The bridge is where the claim “no Earth content” is at its weakest. **SECOND** is an Earth-derived constant sitting in the core crate, and the tick’s **length** is fixed by convention against it, as chapter 2 admitted.

What the design achieves is not the elimination of that dependency — that is not available to anyone — but its **localisation**. There is one constant, in one declared place, named by one identifier, with a lint that fails the build if a second one appears. The arithmetic is clean; the boundary is dirty; and the boundary is visible.

### INTERPRETATION

Whether that is enough is a fair question and this book does not think it is settled. What can be said is that the alternative designs are worse in a specific way: a system with conversion factors distributed through its operations has the same dependency and no place to point at.

A dependency you can point at can be argued about. One that has been dissolved into the arithmetic cannot.

### WHAT THIS CHAPTER ESTABLISHED

- Earth enters through one exact integer constant, **SECOND**, and a lint fails the build if the core names a foreign unit anywhere else.
- Conversion **in** is multiplication and never rounds; conversion **out** is division and is the only place a rounding mode is chosen.
- Input finer than  $10^{-30}$  s is refused, not rounded — the third refusal in three chapters.
- Whole seconds land with 30 trailing base-5 zeros and whole nanoseconds with 21, which makes good invariant tests and explains why UCID has no entropy.
- TT is the only pivot; all 27 leap seconds live at the parse/format boundary, offline. The 1961–1972 rubber-second era turns out exactly representable, by accident.
- The bridge does not eliminate the Earth dependency. It localises it to one place you can point at and argue about.

CHAPTER 8

# 8

## Derived calendars

---

The ladder of chapter 4 has no days and no years in it, deliberately. But people live on planets, and a calendar that cannot say what day it is locally is an instrument of limited interest.

So there is a second layer: local calendars, derived from a body's own motion. This chapter is how that derivation works. Part V is how far it generalises and where it fails.

## Four components

Every calendar in the system is the same shape:

| component | what it is                                                        | source                            |
|-----------|-------------------------------------------------------------------|-----------------------------------|
| Body      | rotation, solar day, orbital period — as exact rationals of ticks | cited measurement                 |
| Anchor    | where the count starts in phase                                   | <b>declared, cited, empirical</b> |
| LeapRule  | intercalation: how the fractional day is absorbed                 | <b>derived</b>                    |
| Cycles    | grouping periods, if the calendar names a satellite               | <b>derived</b>                    |

Three of the four are computed. One is not, and the one that is not is the interesting one — it gets flagged here and gets a full treatment in chapter 16.

## Units, as rationals of ticks

A body’s periods enter as exact rational numbers of ticks, each carrying its citation, its epoch, its secular rate, and the window over which it is valid.

That last part matters more than it sounds. Earth’s rotation is slowing by roughly 1.8 milliseconds per century; its tropical year is shortening by about half a second per century. A parameter treated as a constant is wrong the moment you leave the epoch it was measured at — and this system is aimed at deep time, where “leaving the epoch” is the normal case.

So evaluating a parameter outside its stated window produces a warning rather than a confident extrapolation. The system will tell you the number is out of its depth.

## Intercalation is derived, never declared

Here is the heart of the mechanism.

A body's year is not a whole number of its days. Earth takes about 365.2422 rotations per orbit. Some scheme has to absorb the 0.2422, and every civilisation that has built a calendar has invented one.

This system does not invent one. It **computes** the schemes, by expanding the fractional part as a continued fraction and reading off the convergents — the sequence of best rational approximations, in the precise sense that no fraction with a smaller denominator comes closer.

• • • *ucal cal show earth-d — the derivation*

```
intercalation:
  whole_days_per_year  365
  rule                 31/128
  bound                1 local day per 10000 local years
  walked:
    1: 1/4             - 1 day slips in 128 local years
    2: 7/29            - 1 day slips in 1234 local years
    3: 8/33            - 1 day slips in 4269 local years
    4: 31/128          - 1 day slips in 400000 local years  <- chosen
    5: 752/3105        - 1 day slips in 62100000 local years
    6: 4543/18758      - 1 day slips in 937900000 local years
```

Read the first line of that walk. Convergent 1 is  $1/4$  — one leap day every four years. That is the Julian calendar, derived from Earth's rotation and orbit with no knowledge of Rome.

The system then walks the ladder until a convergent meets the requested drift bound, and reports the entire walk rather than only the winner — so you can see what was rejected and why.

#### INTERPRETATION

The reporting-the-whole-walk decision looks like a nicety and is not. A mechanism that returns only its answer cannot be argued with. One that shows its working can be checked, and — as chapter 10 will describe at length — can turn out to contradict the person who built it.

The single most important output this project ever produced came from reading a walk and noticing what was **not** in it.

## Cycles, and the absence of them

Months are grouping periods, and they come from a satellite.

If a calendar names a grouping satellite, the system derives the cycle structure the same way — continued-fraction expansion of the ratio between the orbital period and the satellite’s synodic period. Earth names the Moon, and the derivation produces 235/19, the Metonic cycle, known since antiquity and recovered here from Earth’s periods alone.

If a calendar names no grouping satellite, it has no months. Not a default month, not a synthesised one, not a fallback to Earth’s. None.

● ● ● *ucal cal list — three derived calendars*

```
earth-d:
  leap_rule      31/128 (convergent 4)
  cycles         from moon

mars-d:
  leap_rule      45/76 (convergent 6)
  cycles         none — the calendar names no grouping satellite

titan-d:
  status         no anchor: complete in units, intercalation and cycles,
                  incomplete in phase. Asking for local fields is UCAL-
E0062.
```

MARS HAS NO MONTH, AND THAT IS THE CORRECT OUTPUT

Mars has two satellites. Phobos orbits in about 0.45 of a Martian day; Deimos in about 5.4. Neither is anything a person would call a month.

The system could have synthesised one — divided the Martian year into twelve, say, or borrowed a period from somewhere. It returns nothing instead, because a month-shaped structure imposed on a body that has no month is Earth structure leaking through a mechanism built specifically to keep it out.

The absence is not a gap in the implementation. It is the implementation working.

## The anchor: the one thing that cannot be derived

Everything above came out of arithmetic on cited parameters. The anchor does not, and cannot.

Intercalation tells you how long a year is. It does not tell you **when the year starts** — and no amount of tick counting will, because phase is not a consequence of period. You can know exactly how long Earth takes to rotate and still not know whether it is currently noon.

So the anchor is declared: one cited, interval-valued constant per body, with a revision number and an uncertainty window.

A calendar with no anchor is not an error to construct. It is an error to **ask local fields of:** `UCAL - E0062`. Titan is in exactly that state, and chapter 16 explains why it will stay there.

The anchor is where this mechanism admits it is not self-sufficient, and the admission is structural rather than apologetic. It is one declared constant, no more privileged than the rotation period sitting beside it and no less necessary.

What would have been dishonest is a default — an anchor that quietly fell back to Earth's, or to zero, so that every body appeared to have a working calendar. That would produce confident local dates for a body whose phase nobody knows.

The Gregorian calendar is in this system, and it is not a derived calendar.

• • • *ucal cal list — the legacy entries*

```

earth-civil:
  kind      legacy – declared tables (§8.6)
  arbitrary  4
  leap_rule 97/400 (NOT a convergent – declared, not derived)

earth-julian:
  kind      legacy – declared tables (§8.6)
  arbitrary  4

```

`earth-civil` has four items of **arbitrary content** — irregular month lengths, the seven-day week (which corresponds to no astronomical period at all), the leap rule, and the epoch. None of them follows from Earth’s motion. They are declared tables, and the type system keeps them in a separate trait: a function that wants a derived calendar will not accept a legacy one, and there is no blanket conversion.

Note the parenthesis on the leap rule. `97/400` is marked **NOT a convergent**. That label is the subject of chapter 10, and it is where this book’s most uncomfortable finding lives.

**INTERPRETATION**

It is worth being explicit that “legacy” here is a technical classification and not a judgement. A declared table is not a worse calendar than a derived one; the Gregorian reform solved a real problem and solved it well enough to still be in use.

What the classification records is **where the authority comes from** — arithmetic on cited parameters, or a decision someone made. Part VI applies that same distinction to scriptural chronologies, and it has to apply it evenhandedly or not at all. A system that calls the Gregorian derived and Seder Olam declared would be doing something other than classifying.

## WHAT THIS CHAPTER ESTABLISHED

- Every calendar is (Body, Anchor, LeapRule, Cycles). Three are computed; the anchor is not.
- Body parameters are exact rationals of ticks carrying epoch, secular rate, and validity window — and evaluating outside the window warns rather than extrapolating.
- Intercalation is derived by continued-fraction expansion, and the whole walk is reported, not just the winner. Convergent 1 for Earth is  $1/4$  — the Julian rule, derived.
- Cycles come from a named grouping satellite or do not exist. Mars has no month, and the absence is the mechanism working.
- The anchor cannot be derived, because phase is not a consequence of period. It is one declared, cited, interval-valued constant, and its absence is `UCAL-E0062` rather than a default.
- Legacy calendars are held apart by the type system. `97/400` is marked **not a convergent** — the subject of chapter 10.

PART III

---

# What implementation refused

*Design as something that loses arguments.*

CHAPTER 9

# 9

## Divergence

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

### **Specified content**

- Which UCAL-1 rules survived contact with the compiler, which were altered, which were dropped, and what each change cost.

- Figure F10 — the divergence map, rules coloured by fate.

CHAPTER 10

# 10

## The 97/400 correction

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 — the centrepiece and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

### Specified content

- Two RFC revisions asserted the machinery reproduces the Julian **and** Gregorian rules as convergents. The Julian rule is convergent 1.

- $97/400$  is not a convergent at any depth.  $8/33$  is more accurate with a denominator twelve times smaller;  $31/128$  is  $124\times$  more accurate.
- The machinery contradicted its author, and the author published the contradiction — dated, including the revisions that were wrong.
- The direct answer to Maimonides' charge in **Guide** I.73 that such systems are built to make their conclusions provable.

PART IV

---

# The datum

*Why it cannot be measured, and what was done instead.*

CHAPTER 11

# 11

## Why it cannot be measured

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

### Specified content

- Rule Q's three reasons: exactness cannot come from measurement; the FLRW  $t \rightarrow 0$  limit is not an observable event; the extrapolation is model-dependent.

- Kant's fourth, which the specification does not state: the question presupposes a completed totality that is not a possible object of knowledge.
- The datum in ordinary company — TAI 1958, the Julian Day epoch, Unix — and the exact parallel to the SI second.

CHAPTER 12

# 12

## What was done instead

---

**NOT YET DRAFTED**

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

**Specified content**

- `BIG_BANG_CLAIM` as a `SignedWindow` with no arithmetic implementations, `UCAL-E0025`, and the compile-fail test proving the type cannot reach an operand position — **the book's central exhibit**.
- `datum_provenance` as machine-readable, re-executable data, with its  $-0.017\ 190\ 364$  s rounding residual.
- Kant's constitutive/regulative distinction: Kant policed the boundary with philosophical discipline; the crate polices it with a type.
- `UC-Θ` as unbuilt — the profile in which the datum is the beginning of time at organization, and what it would cost.

CHAPTER 13

# 13

## Florensky's radius

---

**NOT YET DRAFTED**

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

**Specified content**

- **Мнимости в геометрии** (1922) as the closest precedent and the clearest cautionary tale in one text.

- The Dante geometry as genuine achievement; then §9, where a formal artifact is read as a physical place.
- The distance between the two is exactly one move — and Florensky had no rule against the second.
- Basil, **Hexaameron** I.6, and the historical cost: persecution from 1922 to execution in 1937.

PART V

---

# **Any celestial body**

*How far the approach generalises, and precisely where it  
fails.*

CHAPTER 14

# 14

## Universal baselines

---

**NOT YET DRAFTED**

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

**Specified content**

- Nothing in the arithmetic references a rotation, an orbit, or a civil calendar.

- One derivation mechanism — every calendar is (Body, Anchor, Cycles, LeapRule)  
— with Earth as an ordinary instance and no crate named after a body.

CHAPTER 15

# 15

## Deriving a calendar

---

**NOT YET DRAFTED**

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

**Specified content**

- Earth: convergents  $1/4$ ,  $7/29$ ,  $8/33$ ,  $31/128$  — the Julian rule as convergent 1, the Gregorian absent.

- Mars: 668.59 sols per year, convergent 16/27.
- Titan: tidally locked, handled with no special case — which is the point.
- The Metonic cycle 235/19 derived from Earth's periods unaided.

CHAPTER 16

# 16

## Where it breaks

---

**NOT YET DRAFTED**

This chapter is specified in RFC UCAL-A1 §16.2 — no shorter than ch. 15 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

**Specified content**

- The anchor is empirical and cannot be derived.

- A body with no qualifying satellite has no month — Mars is the worked case, and the absence is the correct output.
- A rogue planet has no year.
- A tidally locked body's day equals its orbit.
- Relativistic environments are out of scope.
- Body parameters carry secular rates and validity windows and are wrong outside them — which matters most at exactly the deep-time scale this project targets.

CHAPTER 17

# 17

## The one qualification

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 — one page and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

### **Specified content**

- For a single planet this is a curiosity. For timekeeping across two or more bodies, a universal ladder with local overlays may be the only coherent arrangement.

- A claim about coherence, not a recommendation for adoption.

PART VI

---

# Nine readings

*The traditions as readers of the artifact — neither  
validating it nor validated by it. Every chapter names its  
conflict.*

CHAPTER 18

# 18

## Greek

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A1 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

### Specified content

- Archimedes' **Sand Reckoner** builds a positional hierarchy to make a cosmic magnitude expressible, arriving at  $\sim 10^{63}$ .

- Euclid X.2's anthyphairesis **is** the intercalation algorithm.
- **Conflict:** Aristotle holds time a continuum and the instant a limit; the instrument is discrete. And **Physics** IV.14 makes the datum's objectivity depend on there being a counter.

CHAPTER 19

# 19

## Jewish

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A2 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

#### **Specified content**

- **Molad tohu** — “the new moon of chaos” — a stipulated epoch placed before the event it anchors, nineteen centuries early.

- The **helek** chosen for divisibility, exactly as **SECOND** was.
- **Conflict:** the compressed Persian period is provenance overruled by doctrine, and the crate's re-executable chain would expose it — so the instrument judges a tradition, which the book's own rules forbid the author from doing.

CHAPTER 20

# 20

## Islamic

---

**NOT YET DRAFTED**

This chapter is specified in RFC UCAL-A1 §17.2 — A3 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

**Specified content**

- Ghazālī’s ‘āda — divine custom yielding practical certainty without necessity — is what a validity window encodes.

- **Conflict:** under Premise 6 an orbital period is a habit, so a guaranteed drift bound over 400,000 years is meaningless. The crate's ontology is Rushdian and it took a side without arguing for it.

CHAPTER 21

# 21

## Patristic and Latin

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A4 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

### Specified content

- Basil’s road-and-house image is Rule Q’s content exactly.
- Augustine’s answer is `UCAL - E0020`: an error, not a negative number.

- **Conflict: *ex nihilo*** is held firmly, so UC- $\Theta$  is heterodox by this standard — and the book must say so rather than soften it.

CHAPTER 22

# 22

## Orthodox

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A5 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

### Specified content

- Gregory of Nyssa’s διάστημα makes a calendar an instrument of the created order by definition — the limitation is the doctrine, not a defect.

- Palamas' essence/energies distinction as the strongest licence the thesis receives from any tradition.
- **Conflict:** Rule N sides with the 1913 Synodal position against Florensky and Losev — the two thinkers this book most depends on.

CHAPTER 23

# 23

## Latter-day Saint

---

**NOT YET DRAFTED**

This chapter is specified in RFC UCAL-A1 §17.2 — A6 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

**Specified content**

- D&C 130:4–5 states Rule K’s thesis in 1843: reckoning is body-relative.
- Abraham 3:4 is a declared bridge constant with a profile tag on each side.

- **Conflict:** Kolob is a governing **body**, where Rule K.5 permits only an abstract ladder — the privilege is located differently.

CHAPTER 24

# 24

## Modern philosophy

---

**NOT YET DRAFTED**

This chapter is specified in RFC UCAL-A1 §17.2 — A7 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

**Specified content**

- The First Antinomy supplies a fourth reason for Rule Q the specification does not state.

- Constitutive versus regulative, enforced by a type with no arithmetic: Kant policed the boundary by discipline, the crate by the compiler.
- **Conflict:** Kant's number is the schema of magnitude — monadic — which contradicts the eidetic number of A1 and A8, and Rule G asserts both readings of one integer.

CHAPTER 25

# 25

## Russian

---

**NOT YET DRAFTED**

This chapter is specified in RFC UCAL-A1 §17.2 — A8 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

**Specified content**

- Fyodorov makes total addressability of the past an obligation.

- Vernadsky's process-relative times and Chizhevsky's solar-period dating as Rule K's unacknowledged ancestry, a century early.
- **Conflict:** Bugaev proposes discreteness **as worldview**, which the book declines — from the closest kin. And Losev's *имя* makes the name constitutive where Rule N makes it a locale-table entry.

CHAPTER 26

# 26

## Method

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A9 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

### **Specified content**

- Klein on **arithmos**; Sorabji on time, creation, and the continuum.

- **Conflict:** the method makes the code the invariant and the traditions the variables. That is itself a metaphysical choice, made without argument — and disclosing it is the last and most self-referential application of Rule M.

PART VII

---

# The instrument as research tool

*What a program can do that an argument cannot: it can  
be run against material its author did not choose, and  
return an answer he did not want.*

CHAPTER 27

# 27

## Six samples

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §18 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

### **Specified content**

- S1 — comparative chronology on one axis: Seder Olam, Byzantine, Ussher, D&C 77, each a declared profile.

- S2 — calendar audit by convergent. Julian **is** convergent 1; Gregorian is not a convergent at any depth; Jalali 8/33 **is** convergent 3.
- S3 — uncertainty audit: cited versus stipulated.
- S4 — cross-body simultaneity: the demonstration that “now” is not a shared object.
- S5 — measuring διάστημα with no Earth content in the units.
- S6 — a revealed ratio evaluated as a bridge constant, refused as an anchor.

CHAPTER 28

# 28

## Null results

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

### **Specified content**

- Each sample's negative finding, stated as plainly as its positive one.

- Where a sample produced nothing, say so; where a result was weaker than hoped, say how much.
- This chapter is what distinguishes the book from a demonstration.

PART VIII

---

# The claim

CHAPTER 29

# 29

## Why a program

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

### **Specified content**

- An essay can assert that a distinction ought to be respected; a type system can make violating it fail to build.

- Why prose cannot be **run** by someone who disagrees.

CHAPTER 30

# 30

## Uselessness restated

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

### **Specified content**

- The thesis, not an apology. Part V's qualification restated as a limit rather than a rescue.

CHAPTER 31

# 31

## What is not claimed

---

### NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

### **Specified content**

- The negative inventory, assembled from Appendix C.

CHAPTER 32

# 32

## Kant's moon

---

**NOT YET DRAFTED**

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

**Specified content**

- Transcendental illusion is not error but a natural appearance that persists after diagnosis — the astronomer knows the moon is not larger at the horizon and still sees it that way.
- Every reader who sees a 61-digit integer will read it as a fact about being, and no rule prevents that.
- **UCAL-E0025** does not cure the illusion; it refuses to compute with it, which is the only thing a specification can do.

AFTERWORD

# About the author

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Vladimir Ulogov has spent decades building infrastructure for distributed systems — the kind of software that watches other software. Early in his career he worked on monitoring and telemetry platforms; later years took him into federated observability, telemetry buses, and the architecture of systems that have to make sense of millions of data points without losing the thread.

Observability, in the end, is a discipline of **coherence** — of never reporting a state the system cannot account for, of insisting that every signal follow from something real. It is not an accident that a calendar built to point at an origin it cannot measure carries the same instinct: it declares what it knows, declares what it has merely stipulated, and refuses to let the second quietly become the first.

What makes him slightly unusual in his corner of the industry is a tendency to write his own tools — not small utilities, but programming languages. The Bund language (its compiler, its VM, its document store, its parser) lives in a long series of Rust crates on crates.io. `rust_dynamic`, `rust_multistackvm`, `bundcore` — each is a building block that exists because the off-the-shelf options didn't fit the shape of the work. `ucal` grew the same way, from an irritation about units that no existing library was going to fix, because no existing library thought it was a problem.

## A work of love

`ucal` is open source, under a permissive licence — you can read it, fork it, study it, modify it, and pass it on. It carries no analytics and no telemetry. It was not built to be sold and it was not built to be adopted; it was built because the question would not go away, and because the only way to find out whether a distinction can be enforced by a compiler is to try to enforce one.

## A note on cooperation

Vladimir believes firmly in the human capacity for mutual help — that we make better work, and live better lives, when we share what we know and what we build. Open source is one of the most concrete expressions of cooperation our era has produced: code read, improved, and passed forward without payment, without permission, by people who will never meet.

This book is offered in the same spirit. If it helps you see one distinction more clearly — between what an instrument measures and what it merely points at — that is enough.

## Where to find more

**GitHub** `@vulogov` — the source for `ucal`, `Bund`, and the dozen-plus Rust crates that carry the infrastructure. Issues and pull requests welcome.

**LinkedIn** `/in/vladimirulogov` — posts on observability, the occasional long-form essay.

**YouTube** `@vulogov` — talks and walkthroughs from the conference trail.

*If you find a claim in this book that the source tree does not support, open an issue. That is the only kind of correction the book is built to receive, and it is the kind it most wants.*

AN INSTRUMENT FOR THE IMMEASURABLE

E N D   O F   T H E   B O O K